

Ownership Intelligence Engine

Master orchestration layer that determines the ownership-control pathway required to convert a probate-related property into an acquisition opportunity.



Strategic Moat: From "Finding Properties" to Ownership-Control Pathways

Most probate data products operate under a lead-generation paradigm: find a name in court records and sell it as a lead. Gieni operates under an acquisition intelligence paradigm. Real estate cannot be purchased from a "probate record"—it can only be acquired by resolving the legal ownership structure, mapping who holds actual disposition control, and establishing an actionable title pathway.



Why This Matters: The Five Questions That Define a Real Deal

Most probate investors buy lists and waste 90% of their outreach time.

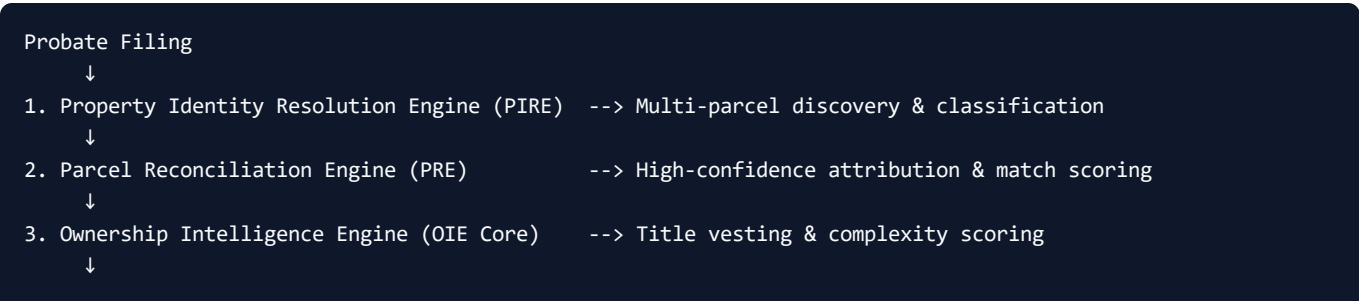
Gieni answers five core questions before an opportunity is ever delivered:

- 1. Which exact property belongs to the estate?
- 2. Who actually controls the disposition?
- 3. How much real, protected equity exists?
- 4. What title or heir complexity must be cured?
- 5. Does probate actually affect the legal ability to sell?

Only after answering all five questions does an opportunity become acquisition-ready.

1. The Master Pipeline Architecture

OIE orchestrates five specialized, patentable-grade submodules that convert raw courthouse filings into verified, decision-maker-ready opportunity files:



4. Control Intelligence Engine (CIE)	--> Ownership ≠ Control pathway separation
↓	
5. Equity Intelligence Engine (EIE)	--> Net equity waterfall & encumbrance audit
↓	
Authority Resolution Engine (ARE)	--> Fiduciary powers & state machine
↓	
Opportunity Scoring Engine (OSE)	--> 0-100 Priority ranking & SLA assignment

2. Submodule 1: Property Identity Resolution Engine (PIRE)

Probate dockets rarely map 1:1 to a single suburban home. A single decedent frequently owns multiple parcels, unbundled acreage, fractional interests, or assets under distinct legal addresses. PIRE answers the initial portfolio discovery questions:

- **Estate Asset Enumeration:** Determines the total count of real property assets owned by the decedent across the county and adjacent jurisdictions.
- **Asset Category Classification:** Distinguishes between Primary Homestead, Rental/Tenant-Occupied Asset, Vacant Investment Land, or Contiguous/Adjacent Tax Parcels.
- **Multi-APN Bundle Resolution:** Identifies cases where a single physical home sits across two separate tax parcels (e.g., house on Lot 1, garage/driveway on Lot 2), preventing catastrophic single-parcel contracts.

3. Submodule 2: Parcel Reconciliation Engine (PRE) & Attribution Score

Gieni rejects absolute claims of "100% match." Instead, PRE delivers high-confidence parcel attribution based on multi-source cross-validation and a proprietary 0–100 Parcel Attribution Score:

Parcel Attribution Score (PAS) Weights:

- Source Agreement (Assessor + Recorder + Court Petition): 30%
- Name Match Confidence (Jaro-Winkler string distance): 25%
- Address Correlation (Situs vs. residential voter/death): 20%
- Title Continuity (Unbroken deed recording history): 15%
- Tax Record Alignment (Current taxpayer of record): 10%

Scoring Thresholds:

[90 - 100]	Verified Parcel Match	--> Certified for Automated Pipeline
[70 - 89]	Probable Parcel Match	--> Delivered with Validation Tag
[< 70]	Manual Review Required	--> Routed to Exception Queue (Tasks & Exceptions)



Evidence Standard: Parcel match confidence must exceed the minimum delivery threshold (PAS ≥ 70) established by Gieni quality standards. Any case below 70 is flagged for human researcher verification.

4. Submodule 3: Ownership Complexity Engine (OCE)

OIE calculates an objective Ownership Complexity Score (10–100) reflecting the title friction required to convey marketable fee-simple title. This score feeds directly into Opportunity Scoring:

- **Score 10 (Minimal Friction):** Single Owner / Sole Fee Simple. Decedent owned 100% of the property. Clear testamentary or statutory path.
- **Score 35 (Low Friction):** Joint Ownership (JTWROS / TBE). Property transfers automatically by operation of law to the surviving co-owner outside of probate.
- **Score 45 (Moderate Friction):** Trust Ownership. Non-probate disposition. Requires locating the trust instrument and validating the Successor Trustee.
- **Score 65 (High Friction):** Multiple Heirs / Tenancy in Common (TIC). Intestate estate with multiple heirs holding fractional undivided interests.
- **Score 85 (Severe Friction):** Multiple Estates / Ancestral Heir Property. Title remains in the name of deceased ancestors, requiring serial probate or quiet title actions.
- **Score 100 (Maximum Friction):** Entity + Trust Stack. Complex layered holding structures (e.g., LLC owned by a Living Trust with a deceased managing member and contested probate).

5. Heir Property Models & Resolution Framework

Heir property represents one of the largest untapped opportunities in distressed acquisitions, but carries high execution risk without workflow segmentation. Gieni classifies heir properties into six structural models and four resolution difficulty tiers:



Heir Property Typology:

1. Single Estate: Clean decedent title, intestate with 2–4 immediate heirs. [Resolution: Easy]
2. Multi-Generation Estate: Title in grandparents; requires tracing lineage through deceased children. [Resolution: Complex]
3. Clouded Title: Unrecorded deeds, wild deeds, or ambiguous boundary descriptions. [Resolution: Specialist Review]
4. Missing Heirs: Known living heirs whose whereabouts are unknown. [Resolution: Moderate]
5. Unknown Heirs: Intestate succession requiring genealogical heir searches and affidavits of heirship. [Resolution: Complex]
6. Multiple Probate Chains: Interlocking probates across multiple counties or states. [Resolution: Specialist Review]

6. Submodule 4: Control Intelligence Engine (Ownership ≠ Control)

This distinction is Gieni's most defensible intellectual property moat. In probate real estate, the legal owner of record is frequently NOT the person who holds the legal power to convey title.



The Ownership vs. Control Matrix:

- Owner: Estate of Decedent --> Control: Court-Appointed Personal Representative (Executor / Administrator)
- Owner: Revocable Living Trust --> Control: Successor Trustee (Bypasses probate court entirely)
- Owner: 6 Surviving Heirs --> Control: Family Consensus Leader / Co-signing Heirs
- Owner: LLC or Partnership --> Control: Managing Member / Surviving General Partner
- Owner: Surviving Joint Tenant --> Control: Sole Living Co-Owner (Executor has zero authority)

By isolating Ownership from Control, OIE establishes the foundational data contract that directly feeds the Authority Resolution Engine (ARE). While OIE solves "What is the property and who holds title?", ARE solves "Who has legal signing authority to sell it?"

7. Submodule 5: Equity Intelligence Engine (EIE)

Simple "estimated equity" models fail because they ignore estate-specific liens and statutory claims. EIE computes an audited Net Equity Waterfall:

Net Equity Waterfall Formula:

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Gross Market Value (Conservative AVM / Assessor baseline)
- (Estimated Open Mortgage Balance via original recording + amortization)
- (Delinquent Real Property Taxes + accumulated penalties & interest)
- (Municipal Code Enforcement, Weed, & Demolition Assessment Liens)
- (Federal Tax Liens & State Department of Revenue Liens)
- (Medicaid Estate Recovery Program [MERP] Priority Statutory Liens)
- (Estimated Statutory Probate Court Fees & Fiduciary Commissions)
- (Recorded Creditor Claims Against the Estate)
=====
= Net Actionable Estate Equity
```

Outputs are categorized into four Equity Tiers: Exceptional Equity (>70% net spread), High Equity (50%–70%), Moderate Equity (30%–50%), and Low/Negative Equity (<30% or underwater), which directly informs cash offer capacity.

8. Ownership Investigation Exceptions Engine

Robust automation requires deterministic exception handling. The Exception Engine automatically halts automated processing and diverts anomalous cases to the Manual Investigation Queue (Tasks & Exceptions

DB) upon encountering seven specific triggers:

- **Conflicting Deeds of Record:** Simultaneous or contradictory deed recordings within 180 days of death.
- **Multiple Non-Contiguous APNs:** Estate holds 3+ distinct properties requiring allocation decisions.
- **Unrecorded Trust Reference:** Deed conveys to a trust but no certification of trust or docket notice is found.
- **Missing Probate / Broken Deed Chain:** Gap between recorded deed grantee and decedent name.
- **Generational Name Collision:** Father and son sharing exact name (Sr./Jr.) without generational suffixes on title.
- **Out-of-State / Foreign Ownership:** Ancillary probate requirements in other state jurisdictions.
- **Dissolved Entity Ownership:** Property titled under an LLC that was forfeited or administratively dissolved prior to death.

9. Complete OIE Delivery Data Contract

Every completed opportunity delivered by Gieni carries an exhaustive, five-dimension property intelligence profile:



OIE Certified Opportunity Package Schema:

- 1. Property Profile: Situs Address, County, APN/PIN, Legal Lot/Block, Living Area, Lot Size, Land Use Code
- 2. Ownership Profile: Current Vesting of Record, Recorded Deed Instrument #, Deed Type, Ownership Structure, Complexity Score (10-100)
- 3. Equity Profile: Estimated Market Value, Assessor Value, Open Mortgages, Tax Delinquency, Lien Exposure, Net Equity Tier
- 4. Authority Inputs: Identified Fiduciaries, Trustees, Surviving Joint Tenants, Primary Potential Signers (Transferred to ARE)
- 5. Risk Profile: Title Risk Rating, Probate Dependency Rating, Transfer Friction, Marketability Score